6/8/25

Digital Signal Processing

⇒ purpose of Laplace Transform : for all 'R' (real numbers) complex numbers.

$x(t) \rightarrow X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$; $s = \sigma + j\omega$
(only for continuous)

⇒ z-transform

$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}}$$; $z = re^{j\omega}$ → phase. [z-transform]

r → magnitude

when $r = 1$ we get fourier transform (i.e, unit circle)

$X(\omega) = \sum_{n} x(n) e^{-j\omega n}$ (discrete time fourier transform (DTFT))

$X(\omega)$ → continuous variable.

→ To make 'ω' aslo discrete in frequency domain;

$\omega \Rightarrow \frac{2\pi k}{N}$

⇒ Expansion of z-transform

$X(z) = \cdots + x(-1) z^{1} + x(0) z^{0} + x(1) z^{-1} + \cdots$

⊛ $X(z) = \sum_{n} x(n) z^{-n}$ tends to ∞ when $z = 0$

$X(z) = \sum_{n} x(n) z^{-n}$ is finite for entire plane except $z = 0$

So, ROC is entire z-plane except $z = 0$ So, '0' is a pole.

⇒ $x(n) = \delta(n-k)$

$X(z) = z^{-k}$ converges for all k.

$\frac{1}{z^{k}}$

Ex :

$x(n) = \{1, 2, \underset{\uparrow}{9}, 3, 0, 1\}$ → origin.

$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$= 1 \cdot z^{+2} + 2 \cdot z + 9 \cdot z^{0} + 3 \cdot z^{-1} + 1 \cdot z^{-3}$

$= z^{2} + 2z + 9 + 3z^{-1} + z^{-3}$

ROC is entire s-plane except $z = 0$ & $z = \infty$

~~origin~~

Ex: 2.

$x(n) = \delta(n+k)$ (impulse signal)

$X(z) = \sum_n x(n) z^{-n}$

$= \sum_n \delta(n+k) z^{-n}$ (when, $n = -k$) it has value.

$= z^{k}$

ROC : entire z-plane except $z = \infty$

⇒ Inverse z-transform

$X(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$

$x(n) = \{ \underset{\uparrow}{1}, 2, 5, 7, 0, 1 \}$

} Inspection Method

*) If the signal is of infinite duration

$x(n) = \left(\frac{1}{2}\right)^n u(n)$

$X(z) = \left[u(0) + \frac{1}{2}u(1) + \left(\frac{1}{2}\right)^2 u(2) + \dots\dots\right] z^{-n}$

$= 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \dots$

$= \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$

Sum of infinite geometric series $= \frac{1}{1-A}$ (if $A < 1$)

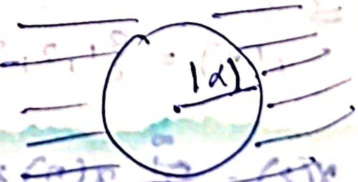
ROC $= z > \frac{1}{2}$

→ ~~Inverse z transform for Infinite:~~

$x(n) = \alpha^n u(n)$

$X(z) = \sum_n \alpha^n u(n) z^{-n}$

$= \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{\alpha}{z}\right)^n = \frac{1}{1 - \alpha z^{-1}}$



ROC $= r > |\alpha|$

⇒ ~~Inverse~~ z-transform

$$X(z) = \sum_n x(n)\, z^{-n} \quad ; \; z = re^{j\omega}$$

$$= \sum_n x(n)\, r^{-n} \cdot e^{-j\omega n} \qquad \because \left|e^{-j\omega n}\right| = 1$$

$$|X(z)| = \left|\sum_n x(n) r^{-n} e^{-j\omega n}\right| \le \sum_n \left|x(n)\, r^{-n} e^{-j\omega n}\right|$$

$$\le \sum_n \left|x(n) r^{-n}\right| \left|e^{-j\omega n}\right|$$

$$\le \sum_n \left|x(n) \cdot r^{-n}\right|$$

$$\le \sum_n |x(n)|\, r^{-n} \text{ must be } < \infty$$

*) **Dirichlet Conditions** : $\boxed{\int_T |x(t)|\,dt < \infty}$

⇒ **problems** :

1. $x(n) = \alpha^n u(-n-1)$

$$\left\| \begin{array}{l} x(n) = \alpha^n \\ x(-n) = \alpha^{-n} \end{array}\right.$$

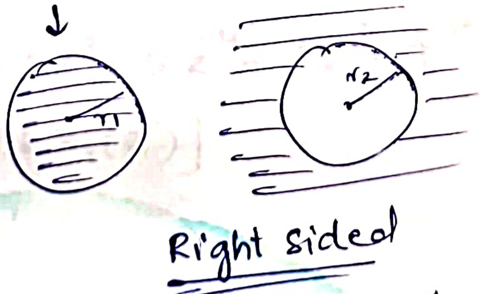
$$X(z) = \sum_{-\infty}^{\infty} \alpha^n u(-n-1)\, z^{-n}$$

$$= \sum_{-\infty}^{-1} \alpha^n z^{-n} \quad ; \text{ take } n = -m$$

$$= \sum_{m=1}^{\infty} (\alpha^{-m})\, z^m$$

2. $x(n) = x_1(n) + x_2(n)$

$x_1(n)$ $\quad$ $x_2(n)$



(*) If when $x_1(n)$ & $x_2(n)$ have ROC individually but when combined ROC will exsists only when $r_1 > r_2$ in this example.

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Linearity.

$x_1(n) \overset{z}{\rightleftharpoons} X_1(z)$

$x_2(n) \rightleftharpoons X_2(z)$

If $x(n) = a x_1(n) + b x_2(n)$

$x(n) \rightleftharpoons X(z)$

$$X(z) = \sum_{n} x(n) z^{-n}$$

$$= \sum [a x_1(n) + b x_2(n)] z^{-n}$$

$$= a \sum x_1(n) z^{-n} + b \sum x_2(n) z^{-n}$$

$$\boxed{X(z) = a X_1(z) + b X_2(z)}$$

prob: $x(n) = [3(2^n) - 4(3^n)] u(n)$

$$a^n x(n) \rightleftharpoons \frac{1}{1 - a z^{-1}}$$

$|z| > |a|$

$$X(z) = \sum_{n} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} (3 \cdot 2^n - 4 \cdot 3^n) z^{-n}.$$

$$= 3 \sum_{n=0}^{\infty} 2^n \cdot z^{-n} - 4 \sum_{n=0}^{\infty} 3^n \cdot z^{-n}$$

$$= \frac{3}{1 - 2z^{-1}} - \frac{4}{1 - 3z^{-1}} \Rightarrow \text{ROC } |z| > 2 \text{ \& ROC } |z| > 3.$$

so, ROC is $|z| > 3$.

$$= \frac{3(1 - 3z^{-1}) - 4(1 - 2z^{-1})}{(1 - 2z^{-1})(1 - 3z^{-1})}$$

$$= \frac{3 - 9z^{-1} - 4 + 8z^{-1}}{(1 - 2z^{-1})(1 - 3z^{-1})}$$

$$= -\frac{(1 + z^{-1})}{(1 - 2z^{-1})(1 - 3z^{-1})}$$

ROC is $|z| > 3$

Time Shifting:

$$x(n) \rightleftharpoons X(z)$$

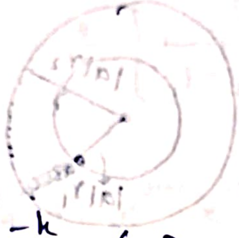
$$\boxed{x(n-k) \rightleftharpoons z^{-k} X(z)}$$

$$X(n-k) = \sum_{-\infty}^{\infty} x(n-k)\, z^{-n}$$

$n-k = m$

$n = m+k.$

$$= \sum_{-\infty}^{\infty} x(m)\, z^{-(m+k)}$$



$$= \sum_{-\infty}^{\infty} x(m) \cdot z^{-m} \cdot z^{-k}.$$

$$= z^{-k} \sum_{-\infty}^{\infty} x(m) z^{-m} = z^{-k} X(z)$$

ROC :

Ex 1

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 5, 7, 0, 1\}$$

$$x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$$

$$x_3(n) = \{\underset{\uparrow}{0}, 0, 1, 2, 5, 7, 0, 1\}$$

$$X_1(z) = z^0 + 2z^{-1} + 5z^{-2} + 7z^{-3} + 1z^{-5}$$
$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

} ROC is entire z plane except $z=0$

$$X_2(z) = z^{+2} + 2z^{+1} + 5 + 7z^{-1} + z^{-3}$$
$$= z^2 + 2z + 5 + 7z^{-1} + z^{-3}.$$

} ROC is entire z-plane except $z = \infty$ & $z = 0$

So, $x_2(n) = x_1(n+2)$ & $X_2(z) = z^2 \cdot X_1(z)$

$x_3(n) = x_1(n-2)$ & $X_3(z) = z^{-2} X_1(z)$

$$X_3(z) = z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$$
$$= z^{-2}[X_1(z)]$$

} ROC is entire z-plane except $z = 0$

Scaling: $x(n) \rightleftharpoons X(z)$ ROC : $r_1 < |z| < r_2$

$$\boxed{a^n x(n) \rightleftharpoons X(a^{-1}z)}$$

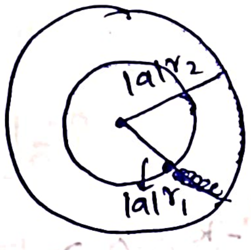
$$X_s(z) = \sum_n a^n x(n) z^{-n}$$

$$= \sum_n x_n (a^{-1}z)^{-n}$$

$$= X(a^{-1}z)$$

$r_1 < |a^{-1}z| < r_2$

$|a|r_1 < |z| < |a|r_2$



ROC $\begin{cases} |a| > 1 \Rightarrow \text{Expands} \\ |a| < 1 \Rightarrow \text{Converge} \end{cases}$

$\rightarrow \; a = r_0 e^{j\omega_0} \quad \Big| \; z = re^{j\omega}$

$a^{-1}z$

$\frac{1}{r_0} \cdot r e^{j(\omega - \omega_0)}$

If $r_0 < 1 \Rightarrow$ Expanding of z-plane
$r_0 > 1 \Rightarrow$ converging of z-plane.

Example:

$$x(n) = a^n (\cos \omega_0 n) u(n)$$

Consider $x'(n) = (\cos \omega_0 n) u(n)$

$$X'(z) = \sum_n (\cos \omega_0 n) u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} (\cos \omega_0 n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} \right) z^{-n}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(e^{j\omega_0 n} + e^{-j\omega_0 n} \right) z^{-n} \quad -①$$

Again consider.

$$x''(n) = (e^{j\omega_0 n}) u(n)$$

$$X''(z) = \sum (e^{j\omega_0})^n u(n) z^{-n} \qquad a^n u(n)$$

$$= \frac{1}{1 - e^{j\omega_0} z^{-1}} \quad -②$$

ROC is $|z| > |e^{j\omega_0}|$

$|z| > 1$

~~from~~ ② Similarly $x'''(n) = (e^{-jw_0 n}) u(n)$

~~① → ½~~

$$X'''(z) = \sum e^{-jw_0 n} \cdot u(n) z^{-n}$$

$$ROC = |z| > |e^{-jw_0}| \qquad = \frac{1}{1 - e^{-jw_0} z^{-1}} \quad -③$$

$$|z| > 1$$

from ② & ③ ① ⇒ $\frac{1}{2}\left[\frac{1}{1 - e^{jw_0} z^{-1}} + \frac{1}{1 - e^{-jw_0} z^{-1}}\right]$

$$X'(z) = \frac{1}{2}\left[\frac{1 - e^{-jw_0} z^{-1} + 1 - e^{jw_0} z^{-1}}{(1 - e^{jw_0} z^{-1})(1 - e^{-jw_0} z^{-1})}\right]$$

$$= \frac{1}{2}\left[\frac{2 - 2(\cos w_0 z^{-1})}{1 - 2\cos w_0 z^{-1} + z^{-2}}\right]$$

1 to ∞

$\boxed{ROC \text{ is } |z| > 1}$

Now $x(n) = a^n (\cos w_0 n) u(n)$

$$X(n) = \sum a^n (\cos w_0 n) u(n) z^{-n}$$

$$= \sum a^n \frac{1 - \cos w_0 z^{-1}}{1 - 2\cos w_0 z^{-1} + z^{-2}}$$

$$= \frac{1}{1 - a\left(\frac{1 - \cos w_0 z^{-1}}{1 - 2\cos w_0 z^{-1} + z^{-2}}\right)^{-1}}$$

$$= \frac{1}{1 - \frac{a - 2a\cos w_0 z^{-1} + az^{-2}}{1 - z^{-1}\cos w_0}}$$

$z \to a^{-1} z$

$z^{-1} \to a z^{-1}$

$z^{-2} \to a^{-2} z^{-2}$

a to ∞

$|z| > |a|$

$r_1 < |z| < r_2$

So, $|z| > r_1$

$|z| > 1$

* $r_1 = 1$

$r_2 = \infty$

prob : $x(n) = -a^n u(-n-1)$

$= \cancel{a^n u(n+1)}$

(E) —

$$X(z) = \sum_n x(n)\, z^{-n}$$

$$= \sum_n -a^n u(-n-1)\, z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} a^n z^{-n} \qquad n = -m$$

$$= -\sum_{m=1}^{\infty} a^{-m} z^m$$

$$= -\left[\sum_{m=0}^{\infty} a^{-m} z^m - 1\right]$$

$$= 1 - \sum_{m=0}^{\infty} a^{-m} z^m \Rightarrow 1 - \sum_{m=0}^{\infty} (a^{-1}z)^m$$

$$= 1 - \frac{1}{1-a^{-1}z}$$

ROC => $|a^{-1}z| < 1$

$|z| < |a|$

$\cancel{\dfrac{a^{-1}z-1}{a^{-1}z}}$ $\cancel{\dfrac{1-a^{-1}z}{a^{-1}z}}$ $\cancel{\dfrac{1}{a^{-1}z}}$

=>

$1 - \dfrac{1}{1-a^{-1}z}$ is same as $\dfrac{1}{1-az^{-1}}$

$$\frac{1 - a^{-1}z - 1}{1-a^{-1}z} = \frac{-a^{-1}z}{1-a^{-1}z} \times \frac{a z^{-1}}{a z^{-1}}$$

$$= \frac{-1}{a z^{-1} - 1} = \frac{1}{1-az^{-1}}$$

$1 - \dfrac{1}{2z} > 0$

$\dfrac{1}{2z} < 1$

$|z| > \dfrac{1}{2}$

11/8/25 ⇒ Practical Applications of z-Transform

1. Convolution : $X(z) = \sum x(n) z^{-n}$.

$x(n) * h(n) \rightleftharpoons X(z)\,H(z)$

2. LTI Systems : poles & zeroes

3. Solving difference : $y(n) = \sum_k b_k y(n-k) + \sum_l a_l x(n-l)$

4. Control System : Design & Analysis.

⇒ Time Reversal :

$x(n) \overset{z}{\rightleftharpoons} X(z)$

$x(-n) \overset{z}{\rightleftharpoons} X(z^{-1})$

$= \sum x(-n) z^{-n}$

$\sum x(m) (z^{-m})^{-1}$

$= X(z^{-1})$

prob : $x(n) = u(-n)$

$X(z) = \sum_{-\infty}^{\infty} u(-n) z^{-n}$

$= \sum_{-\infty}^{0} z^{-n}$ $\quad n = -m$

$= \sum_{0}^{\infty} z^{m} = \sum_{0}^{\infty} (z^{-1})^{-m}$ ~~X(z^{-1})~~

$= \frac{1}{1-z}$ $\quad \boxed{|z| < 1 \text{ : ROC}}$

$X(z) = \frac{1}{1-z^{-1}}$ $\quad |z| > 1$

$X(z^{-1}) = \frac{1}{1-z}$ $\quad |z| < 1$

$r_1 < |z| < r_2$

$r_1 < |z^{-1}| < r_2$

$\frac{1}{r_2} < |z| < \frac{1}{r_1}$ → ROC for $x(-n)$

Ex : $x_1(n) = \begin{cases} \left(\frac{1}{2}\right)^n ; & n \geq 0 \\ \left(\frac{1}{3}\right)^{-n} ; & n < 0 \end{cases}$ find $x_2(n) = x_1(-n)$:

$x_2(n) = \begin{cases} 2^n ; & n \leq 0 \\ \left(\frac{1}{3}\right)^n ; & n > 0 \end{cases}$

$x_1(n) = x_{11}(n) + x_{12}(n)$

$X_{11}(z) = \sum \left(\frac{1}{2}\right)^n u(n) z^{-n}$

$= \sum_{0}^{\infty} 2^{-n} z^{-n}$

$= \frac{1}{1-\frac{1}{2}z^{-1}}$; $|z| > \frac{1}{2}$.

$X_{12}(z) = \sum_{-\infty}^{\infty} \left(\frac{1}{3}\right)^{-n} u(-n-1) z^{-n}$

$= \sum_{-\infty}^{-1} 3^n z^{-n}$

$= \frac{-1}{1-3z^{-1}}$; $|z| < 3$

$$X_1(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{1}{1-3z^{-1}} \; ; \; \frac{1}{2} < |z| < 3 .$$

$$X_2(z) = X_1(z^{-1})$$

$$= \frac{1}{1-\frac{1}{2}z} - \frac{1}{1-3z} \; ; \; ROC = \frac{1}{3} < |z| < 2$$

method: 2 (without properties)

$$x_2(n) = \begin{cases} 2^n \; ; & n \le 0 \\ 3^{-n} \; ; & n > 0 \end{cases}$$

$$x_2(n) = \begin{cases} 2^n u(-n-1) + \delta(n) \\ 3^{-n} u(n) - \delta(n) \end{cases}$$

$$X_2(z) = \sum (2^n u(-n-1) + \delta(n)) z^{-n} + \sum [3^{-n} u(n) - \delta(n)] z^{-n}$$

$$= \frac{-1}{1-2z^{-1}} + 1 + \frac{1}{1-3^{-1}z^{-1}} - 1$$

$|z| < 2$ and $|z| > \frac{1}{3}$; $\frac{1}{3} < |z| < 2$

Verification

Take

$$\frac{-1}{1-2z^{-1}} + 1$$

$$\frac{-1+1-2z^{-1}}{1-2z^{-1}}$$

$$= \frac{-2z^{-1}}{1-2z^{-1}} \times \frac{-2^{-1}z}{-2^{-1}z}$$

$$= \frac{1}{1-\frac{1}{2}z}$$

$$\frac{1}{1-3^{-1}z^{-1}} - 1$$

$$\frac{1-1+3^{-1}z^{-1}}{1-3^{-1}z^{-1}}$$

$$= \frac{3^{-1}z^{-1}}{1-3^{-1}z^{-1}} \times \frac{3z}{3z}$$

$$= \frac{1}{3z-1} = \frac{-1}{1-3z}$$

So, $\boxed{X_2(z) = \frac{1}{1-\frac{1}{2}z} - \frac{1}{1-3z}}$ ROC: $\frac{1}{3} < |z| < 2$

Both methods give same ans.

$\Rightarrow$ prob : $-a^n u(-n-1) \rightleftharpoons \frac{1}{1-az^{-1}}$; $|z| < |a|$: ~~using~~

prove it by using time reversal property

WKT :

sol : $a^n u(n) \rightleftharpoons \frac{1}{1-az^{-1}}$; $|z| > |a|$

Similarly $\left(\frac{1}{a}\right)^n u(n) \rightleftharpoons \frac{1}{1-\frac{1}{a}z^{-1}}$; $|z| > \left|\frac{1}{a}\right|$

consider as $x(n)$

Now $x(-n) = a^n u(-n)$

$$X(z) = \sum_{-\infty}^{+\infty} a^n u(-n) z^{-n} = \frac{1}{1-\frac{1}{a}z} \; ; \; |z| < a.$$

we can write $-a^n . u(-n-1)$ as $-[a^n u(-n) - \delta(n)]$,

so, take $\delta(n) - a^n u(-n)$ as $x_2(n)$

$$X_2(z) = \sum [\delta(n) - a^n u(-n)] z^{-n}$$

$$= 1 - \frac{1}{1-\frac{1}{a}z}$$

$1 - \frac{1}{a}z > 0$

$\frac{1}{a}z < 1$

$z < |a|$

$$= \frac{-\frac{1}{a}z}{1-\frac{1}{a}z} \frac{(-az^{-1})}{(-az^{-1})} = \frac{1}{1-az^{-1}}$$

$\Rightarrow$ Differentiation in z domain.

If $x(n) \rightleftharpoons X(z)$

then $\boxed{n x(n) \rightleftharpoons -z \frac{d}{dz} X(z)}$

proof : $X(z) = \sum_n x(n) z^{-n}$

$$\frac{d}{dt} X(z) = \sum_n x(n) \frac{d}{dt} z^{-n}$$

$$\cancel{\frac{d}{dt}[\sum]} = \frac{d}{dt}\left[\sum_n x(n) z^{-n}\right]$$

$$\sum_n -n x(n) z^{-n-1} = \frac{-1}{z} \sum_n n x(n) z^{-n}$$

take z^{-1} out

Prob: $x(n) = na^n u(n)$

Sol: Let $x_1(n) = a^n u(n)$

then $X_1(z) = \frac{1}{1-az^{-1}}$; $|z| > |a|$

Now $n x_1(n) = x_2(n) = na^n u(n)$

$X_2(z) = -z \frac{d}{dz} X(z)$ (from property)

$= -z \left[\frac{-1}{(1-az^{-1})^2}\right] \left(-az^{-2}(-1)\right)$

$= \frac{az^{-1}}{(1-az^{-1})^2}$; ROC: $|z| > |a|$

ROC is not affected by diffentiation property.

Prob: $X(z) = \log(1+az^{-1})$; $|z| > |a|$. Find the corresponding $x(n)$

Sol: $\frac{d}{dz}(X(z)) = \frac{1}{1+az^{-1}}(-az^{-2})$

$-z\frac{d}{dz}X(z) = \frac{az^{-1}}{1+az^{-1}}$

By using derivative property.

$Z\{n x(n)\} = Z^{-1}\left\{\frac{az^{-1}}{1+az^{-1}}\right\}$

lets take the denominator : $Z^{-1}\left\{\frac{1}{1+az^{-1}}\right\} = (-a)^n u(n)$

Now, $Z^{-1}\left\{\frac{a}{1+az^{-1}}\right\} = a(-a)^n u(n)$

Now, $Z^{-1}\left\{\frac{az^{-1}}{1+az^{-1}}\right\} = a(-a)^{n-1} u(n-1)$

$n x(n) = a(-a)^{n-1} u(n-1)$

$a^{n-1} \cdot a$

$x(n) = \frac{(-1)^{n-1} a^n u(n-1)}{n}$